

COURSE: CS 371 — Advanced Software Engineering

Credits: 3 | Department: Computer Science

Document ID: CAT-COURSE-CS371

Source: <https://catalog.techstate.edu/courses/cs371>

Date Accessed: March 29, 2025

PREREQUISITES: CS 240 AND CS 260 (minimum grade C- for CS 260)

CO-REQUISITES: None

OFFERED: Fall, Spring

DESCRIPTION:

Advanced topics in software engineering. Topics include microservices architecture, API design (REST, GraphQL), event-driven systems, software architecture patterns (CQRS, event sourcing, hexagonal architecture), DevOps practices (CI/CD pipelines, infrastructure as code), observability (logging, metrics, tracing), and large-scale software maintenance.

GRADING: Architecture assignments 40%, Team project (build a microservices system) 40%, Final presentation 20%.